

03

Hands-on Practice

나만의 ML 모델 만들고 웹서비스로 배포하기

Process

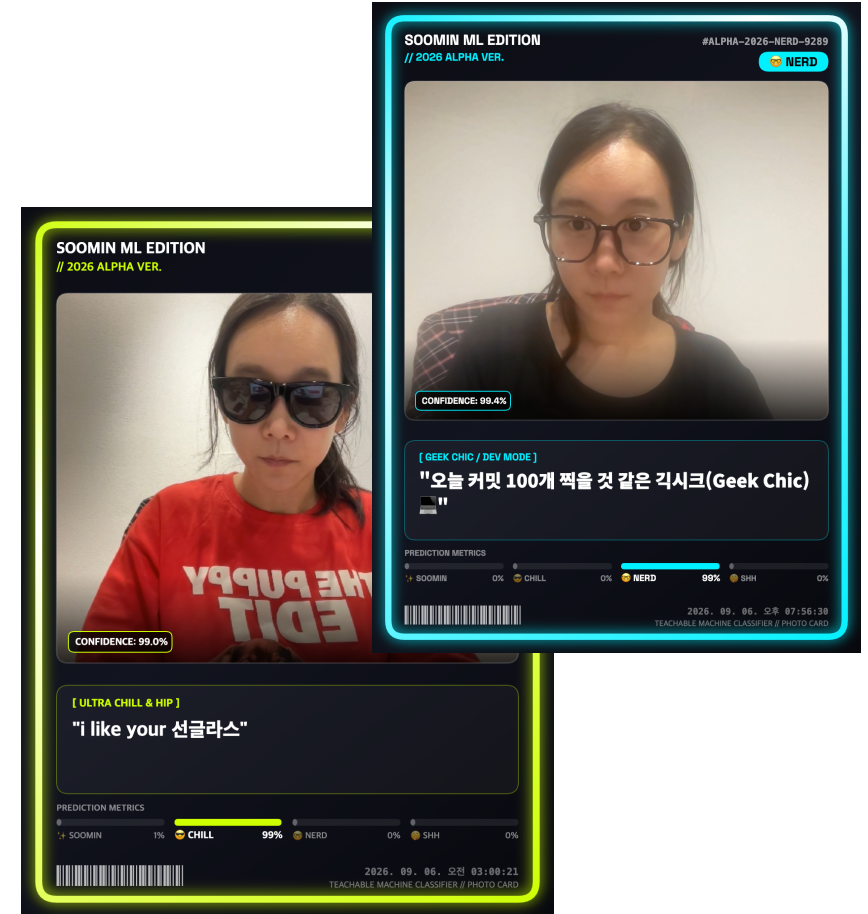
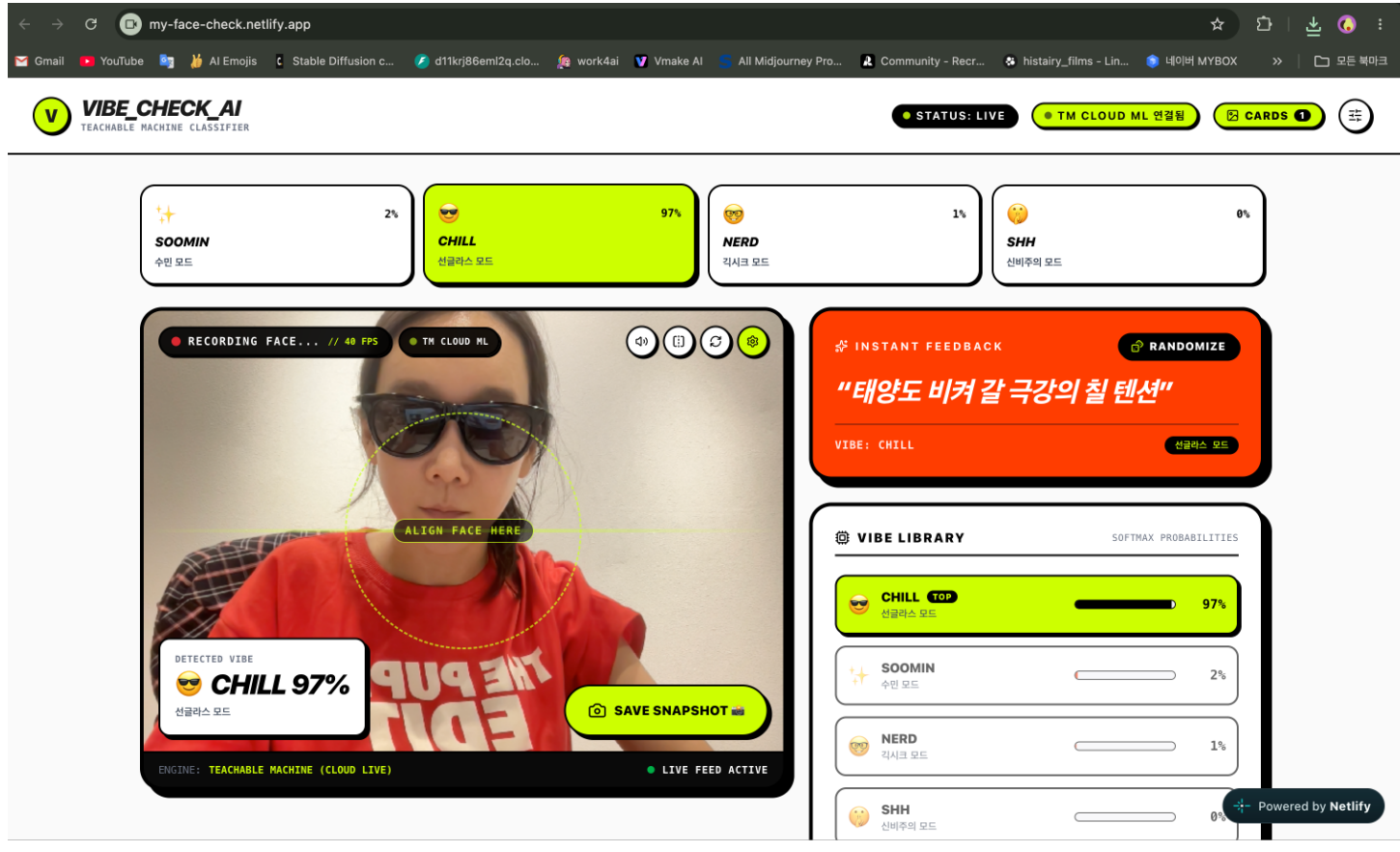
<https://my-face-check.netlify.app/>



구글 계정을 준비해주세요!

Final Output

<https://my-face-check.netlify.app/>



Teachable Machine

- 구글에서 만든 웹기반 노코드 인공지능 학습 툴 (이미지, 사운드, 자세를 인식)
- 링크: <https://teachablemachine.withgoogle.com/>

1. Teachable Machine 웹사이트 접속 (Google 에서 Teachable Machine 검색)



Teachable Machine

Train a computer to recognize your own images, sounds, & poses.

A fast, easy way to create machine learning models for your sites, apps, and more – no expertise or coding required.

Get Started

2. 시작하기 버튼 누르기

Tree 100%

Wings 0%

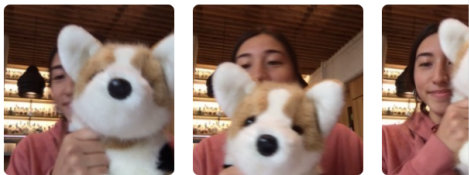
TensorFlow.js ml5.js p5.js Coral Node.js ARDUINO

Teachable Machine

새 프로젝트

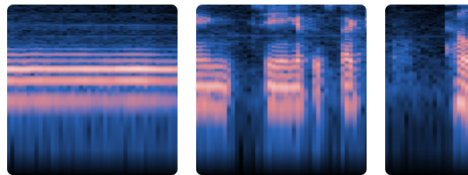
 Drive에 있는 기존 프로젝트를 엽니다.

 파일에서 기존 프로젝트를 엽니다.



이미지 프로젝트

파일 또는 웹캠에서 가져온 이미지를 기반으로 학습시키세요.



오디오 프로젝트

파일 또는 마이크에서 가져온 1초 분량의 사운드를 기반으로 학습시키세요.



포즈 프로젝트

파일 또는 웹캠에서 가져온 이미지를 기반으로 학습시키세요.

3. 이미지 프로젝트 선택

Teachable Machine



Teachable Machine

5. 첫번째 클래스 이름 지정

The screenshot shows the Teachable Machine interface. At the top, a red box labeled '5. 첫번째 클래스 이름 지정' (5. First class name designation) has an arrow pointing to the 'Soomin' class name. Below the class name, there is a section '이미지 샘플 추가:' (Add image samples:) with two buttons: '웹캠' (Webcam) and '업로드' (Upload). A red arrow points to the '웹캠' button, and a red box labeled '6. 웹캠 선택' (6. Webcam Selection) is positioned next to it. Below this, there is another class named 'Class 2' with a similar '이미지 샘플 추가:' section. At the bottom, there is a dashed box with the text '클래스 추가' (Add class). On the right side, there is a '미리보기' (Preview) section with a '모델 학습시키기' (Train model) button and a '고급' (Advanced) dropdown menu. A '모델 내보내기' (Export model) button is also visible.

Teachable Machine

The screenshot displays the Teachable Machine web interface. At the top, the user's name 'Soomin' is shown. The main workspace is divided into two panels. The left panel, titled 'Webcam', shows a live video feed of a person wearing a pink cartoon mask. Below the feed is a blue button labeled '길게 눌러서 녹화하기' (Long press to record) and a settings gear icon. The right panel, titled '21 이미지 샘플' (21 image samples), shows a grid of 21 small images of the person with the mask. A red arrow points from the '길게 눌러서 녹화하기' button to a red instruction box. The instruction box contains the text: '7. 녹화버튼 길게 눌러서 이미지 샘플링 (이미지 직접 업로드도 가능)' (7. Long press the record button for image sampling (image direct upload is also possible)). Below the main workspace, there is a 'Class 2' section with '이미지 샘플' (Image samples) and buttons for '웹캠' (Webcam) and '업로드' (Upload). At the bottom, there is a dashed box with the text '클래스 추가' (Add class).

Soomin

Webcam

21 이미지 샘플

길게 눌러서 녹화하기

Class 2

이미지 샘플

웹캠

업로드

클래스 추가

학습

모델 학습시키기

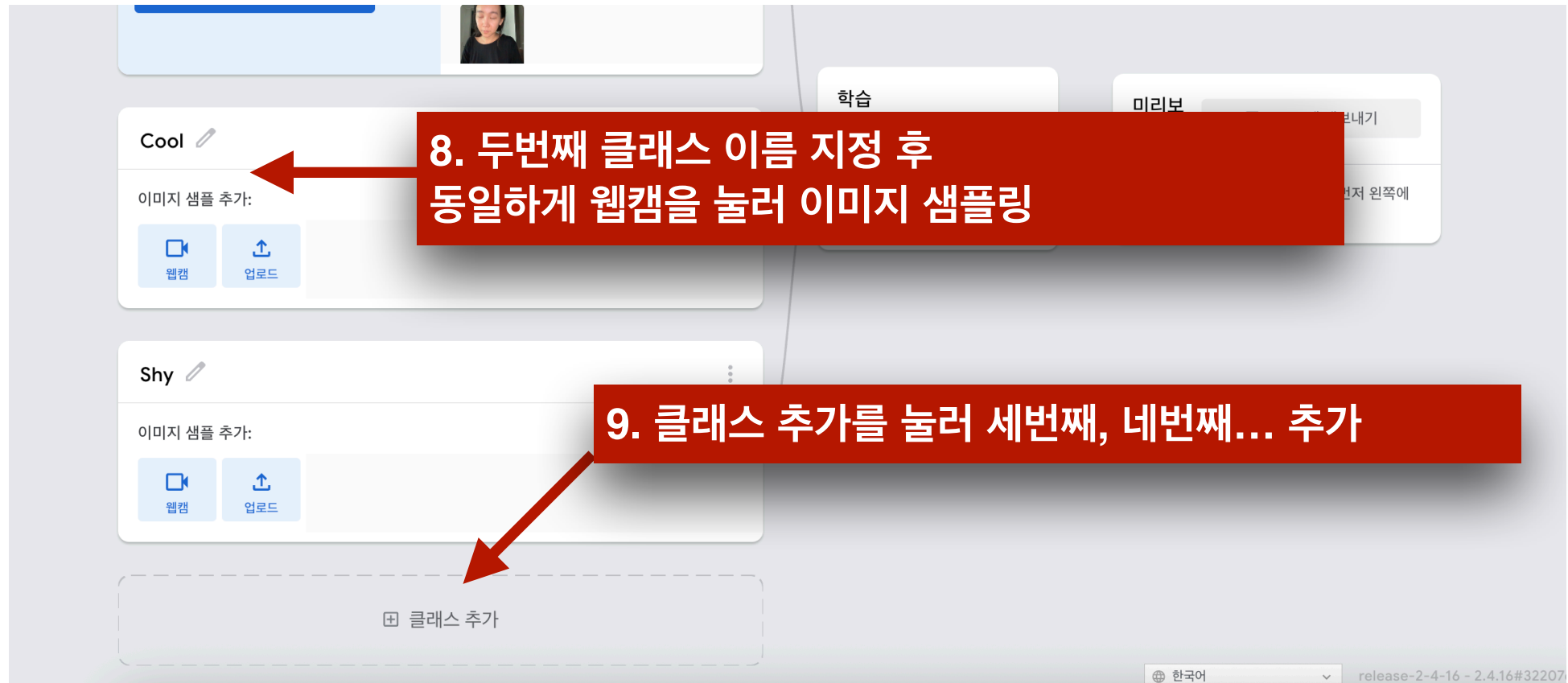
고급

미리보기

모델 내보내기

여기에서 모델을 미리 확인하려면 먼저 왼쪽에서 모델을 학습시켜야 합니다.

Teachable Machine



Teachable Machine

Soomin

21 이미지 샘플

Cool

87 이미지 샘플

Shy

Webcam 122 이미지 샘플

학습

모델 학습시키기

고급

미리보기

모델 내보내기

여기에서 모델을 미리 확인하려면 먼저 왼쪽에서 모델을 학습시켜야 합니다.

탭을 전환하지 마세요.
모델을 학습시키려면 이 탭이 열려 있어야 합니다.

[다시 표시하지 않음](#) [확인](#)

10. 클래스별로 데이터 입력이 완료되면 “모델 학습시키기” 버튼 누르기

Teachable Machine

The interface displays three training classes on the left:

- Soomin**: 21 이미지 샘플 (21 image samples). Includes '웹캠' (Webcam) and '업로드' (Upload) buttons.
- Cool**: 87 이미지 샘플 (87 image samples). Includes '웹캠' (Webcam) and '업로드' (Upload) buttons.
- Shy**: 122 이미지 샘플 (122 image samples). Includes '웹캠' (Webcam) and '업로드' (Upload) buttons.

In the center, a '학습' (Training) section shows '모델 학습 완료됨' (Model training completed) and a '고급' (Advanced) dropdown.

On the right, the '미리보기' (Preview) section shows a live webcam feed. Below it, the '출력' (Output) section displays three bars representing the model's predictions:

- Soomin**: Orange bar
- Cool**: Pink bar
- Shy**: Purple bar with 99% confidence

A red arrow points from the 'Shy' class to the 'Shy' output bar.

11. 미리보기 화면과 출력에서 모델 잘 작동하는지 확인 (혹시 제대로 작동하지 않으면 앞 단계 수정해보세요.)

Teachable Machine

Soomin

21 이미지 샘플

웹캠

업로드

Cool

87 이미지 샘플

웹캠

업로드

Shy

122 이미지 샘플

웹캠

업로드

클래스 추가

12. 모델 내보내기 버튼 누르기

↓

학습

모델 학습 완료됨

고급

미리보기

모델 내보내기

입력 ☐ 사용 Webcam

출력

Soomin

Cool

Shy 99%

Teachable Machine

The screenshot shows the Teachable Machine web interface. In the background, there are three project cards: 'Soomin' with 21 image samples, 'Cool' with 87 image samples, and 'Shy' with 122 image samples. Each card has '웹캠' (Webcam) and '업로드' (Upload) buttons. A modal window is open in the center, titled '프로젝트에서 모델을 사용하려면 모델을 내보내세요.' (To use the model in the project, export the model.). The modal has three tabs: 'Tensorflow.js', 'Tensorflow', and 'Tensorflow Lite'. Under the 'Tensorflow.js' tab, there are three radio buttons: '업로드(공유 가능한 링크)' (selected), '다운로드', and '모델 업로드'. A red arrow points to the '모델 업로드' button. Below the radio buttons, there is a text input field for a shareable link, containing 'https://teachablemachine.withgoogle.com/models/...'. A red banner with white text '13. 모델 업로드 버튼 누르기' (Press the model upload button) is overlaid on the modal. At the bottom of the modal, there is a code snippet for embedding the model into a web page, with a '복사' (Copy) button.

프로젝트에서 모델을 사용하려면 모델을 내보내세요.

Tensorflow.js Tensorflow Tensorflow Lite

모델 내보내기:

☒ 업로드(공유 가능한 링크) ☐ 다운로드

공유 가능한 링크:

모델을 업로드하면 Teachable Machine에서 이 링크에 모델을 호스팅합니다. (FAQ: 내 모델을 사용할 수 있는 사용자는 누구인가요?)

13. 모델 업로드 버튼 누르기

Learn more about how to use the code snippet on [github](#).

```
<div>Teachable Machine Image Model</div>
<button type="button" onclick="init()">Start</button>
<div id="webcam-container"></div>
<div id="label-container"></div>
<script src="https://cdn.jsdelivr.net/npm/@tensorflow/tfjs@latest/dist/tf.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/@teachablemachine/image@latest/dist/teachablemachine-
image.min.js"></script>
<script type="text/javascript">
  // More API functions here:
  // https://github.com/googlecreativelab/teachablemachine-community/tree/master/libraries/image

  // the link to your model provided by Teachable Machine export panel
  const URL = "my_model.js";
```

복사

클래스 추가

Teachable Machine

Soomin

프로젝트에서 모델을 사용하려면 모델을 내보내세요.



모델 내보내기

'모델 업로드'를 클릭하면 모델이 어디에 게시되나요?



모델은 **Google** 서버에 게시되지만 모델을 만들기 위해 사용한 예시는 게시되지 않습니다. 내가 어떤 클래스를 보여 주는지 예측하는 수학 프로그램인 모델만 게시됩니다.

학습 결과물인 모델 파일(가중치 등)만
Google 서버에 게시됩니다.

122 이미지 샘플



웹캠



업로드

```
<div>Teachable Machine Image Model</div>  
<button type="button" onclick="init()">Start</button>
```

```
// https://github.com/googlecreativelab/teachablemachine-community/tree/master/libraries/image
```

```
// the link to your model provided by Teachable Machine export panel
```

```
const URL = "my_model";
```

클래스 추가

Teachable Machine

프로젝트에서 모델을 사용하려면 모델을 내보내세요.

Tensorflow.js | Tensorflow | Tensorflow Lite

모델 내보내기:

☒ 업로드(공유 가능한 링크) ☐ 다운로드 클라우드 모델 업데이트

공유 가능한 링크:

<https://teachablemachine.withgoogle.com/models/5-ImgrBv7/> 복사

모델을 업로드하면 Teachable Machine에서 이 링크에 모델을 호스팅합니다. (FAQ: 내 모델을 사용할 수 있는 사용자는 누구인가)

✓ 클라우드 모델이 최신 상태입니다.

모델에서 사용할 코드 스니펫:

Javascript | p5.js

Learn more about how to use the code snippet on

```
<div>Teachable Machine Image Model</div>
<button type="button" onclick="init()">Start</button>
<div id="webcam-container"></div>
<div id="label-container"></div>
<script src="https://cdn.jsdelivr.net/npm/@tensorflow/tfjs@latest/dist/tf.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/@teachablemachine/image@latest/dist/teachablemachine-
image.min.js"></script>
<script type="text/javascript">
  // More API functions here:

```

복사

클래스 추가

14. 해당 URL을 꼭 메모장에 저장해주세요!

model.json

```
▼ layers:
  ▼ 0:
    class_name: "Dense"
    ▼ config:
      units: 100
      activation: "relu"
      use_bias: true
      ▶ kernel_initializer: { class_name: "VarianceScaling", config: {...} }
      ▶ bias_initializer: { class_name: "Zeros", config: {} }
      kernel_regularizer: null
      bias_regularizer: null
      activity_regularizer: null
      kernel_constraint: null
      bias_constraint: null
      name: "dense_Dense1"
      trainable: true
      ▶ batch_input_shape: [ null, 1280 ]
      dtype: "float32"
  ▼ 1:
    class_name: "Dense"
    ▼ config:
      units: 3
      activation: "softmax"
      use_bias: false
      ▶ kernel_initializer: { class_name: "VarianceScaling", config: {...} }
      ▶ bias_initializer: { class_name: "Zeros", config: {} }
      kernel_regularizer: null
      bias_regularizer: null
      activity_regularizer: null
```

앞 단계에서 추출된 1,280개 이미지 특징을
100개의 내부 특징으로 변환하고,
ReLU를 적용해 활성화된 특징으로 변환

- ReLU는 신경망 중간에 들어가는 아주 단순한 계산 규칙
 $\text{ReLU}(x) = \max(0, x)$

ReLU를 거친 100개 활성화된 특징을 받아
3개의 클래스 점수(Soomin, Cool, Shy)로
변환하고, Softmax를 적용해 확률 형태 변환

- Softmax: 합이 1이 되도록 바꿔주는 함수

weights.bin

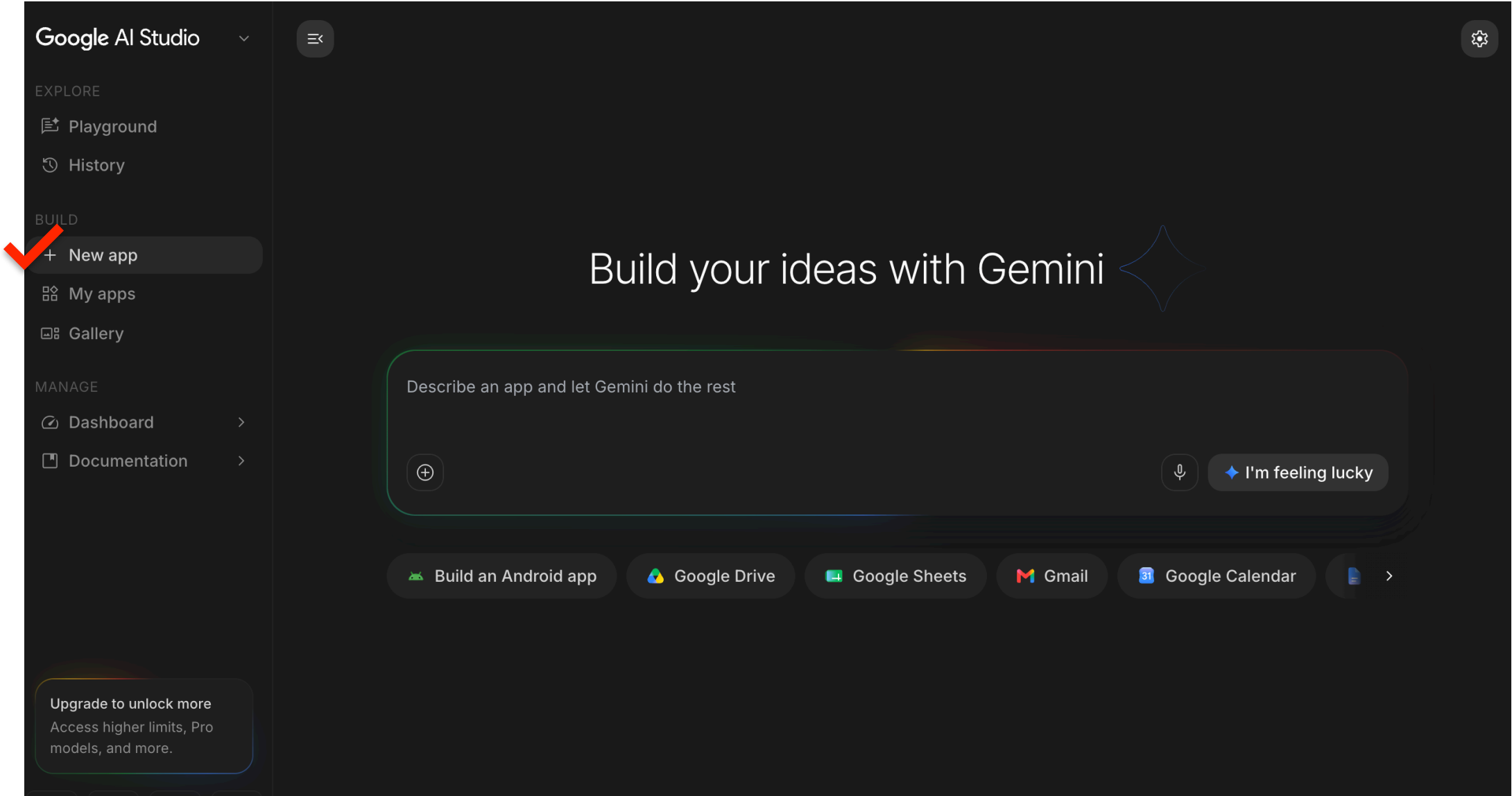
```
weights.bin 를 사람이 읽을 수 있는 숫자로 변환한 파일
총 538608개의 숫자 (float32), 263개의 덩어리(텐서)로 구성
=====
#0 Conv1/kernel [모양 3x3x3x16, 432개]
+0.053619 -0.028929 -0.161487 -0.000000 +0.185963 +0.000000 +0.098091 +0.000000
-0.042291 -0.139291 -0.044071 +0.000000 -0.174370 +0.000000 -0.061369 +0.033705
+0.109852 +0.005884 -0.279647 -0.000000 +0.413372 -0.000000 -0.030402 -0.000000
-0.054262 +0.241658 +0.157489 -0.000000 -0.307682 +0.000000 -0.108046 +0.088464
+0.033152 -0.077727 -0.071000 -0.000000 +0.105327 -0.000000 -0.082935 -0.000000
-0.008772 +0.039734 -0.109442 +0.000000 -0.099733 -0.000000 -0.002816 +0.012401
-0.241117 +0.056381 -0.226949 -0.000000 -0.176207 +0.000000 +0.372518 +0.000000
-0.012842 -0.061853 -0.248377 +0.000000 -0.250394 +0.000000 +0.210416 +0.003359
-0.469431 +0.133549 -0.417099 -0.000000 -0.341784 +0.000000 -0.147311 -0.000000
-0.034969 +0.225201 +0.533734 +0.000000 -0.436903 +0.000000 +0.390539 -0.049633
-0.138797 +0.082366 -0.106949 -0.000000 -0.084892 -0.000000 -0.242123 -0.000000
-0.034464 -0.118882 -0.281396 +0.000000 -0.108728 -0.000000 +0.116050 +0.004875
+0.194826 +0.196980 -0.082316 -0.000000 +0.013497 +0.000000 +0.332034 +0.000000
+0.026093 +0.162733 -0.193251 +0.000000 -0.090098 +0.000000 +0.035395 -0.004692
+0.386667 +0.234532 -0.174614 -0.000000 +0.031299 -0.000000 -0.087632 -0.000000
-0.055090 -0.088768 +0.383678 +0.000000 -0.141520 +0.000000 +0.023823 +0.012517
+0.084210 +0.158713 -0.045223 -0.000000 -0.007567 -0.000000 -0.256965 -0.000000
+0.015032 -0.063764 -0.188668 +0.000000 -0.047392 -0.000000 +0.021561 +0.005018
+0.054503 -0.010599 +0.055929 -0.000000 +0.254367 +0.000000 +0.296852 +0.000000
-0.178889 -0.049701 -0.152819 +0.000000 +0.237529 +0.000000 -0.051054 -0.246081
+0.125518 +0.017815 +0.174718 -0.000000 +0.474715 -0.000000 -0.055262 -0.000000
-0.285701 +0.060948 +0.361078 -0.000000 +0.538439 +0.000000 -0.093907 -0.251884
+0.035451 -0.017107 -0.044384 -0.000000 +0.115069 -0.000000 -0.255073 -0.000000
-0.122441 +0.018279 -0.203820 +0.000000 +0.135405 -0.000000 -0.009428 -0.054587
-0.244933 -0.013695 +0.055855 -0.000000 -0.200639 +0.000000 +0.586538 +0.000000
-0.182382 +0.007881 -0.420639 +0.000000 +0.357026 +0.000000 +0.255047 -0.160656
-0.444985 -0.047392 +0.199176 -0.000000 -0.359758 +0.000000 -0.201910 -0.000000
-0.353187 +0.104339 +0.007337 +0.000000 +0.712344 +0.000000 +0.448483 -0.101416
-0.126168 +0.026857 +0.045101 -0.000000 -0.088594 +0.000000 -0.399620 -0.000000
-0.115607 -0.078802 -0.409917 +0.000000 +0.162582 +0.000000 +0.121630 -0.015718
+0.221572 +0.067123 +0.046381 -0.000000 +0.018099 +0.000000 +0.284987 +0.000000
-0.015380 +0.276914 -0.161954 +0.000000 +0.177308 +0.000000 +0.047400 +0.058268
+0.387601 +0.006272 +0.098265 -0.000000 +0.050319 +0.000000 -0.069652 -0.000000
-0.073691 -0.202629 +0.338170 +0.000000 +0.224644 +0.000000 +0.039774 -0.001271
+0.006119 +0.017916 +0.027222 -0.000000 -0.001025 -0.000000 -0.208575 -0.000000
-0.029370 -0.059514 -0.190525 +0.000000 +0.074523 +0.000000 +0.027381 -0.004920
-0.003084 +0.004183 +0.111438 -0.000000 +0.102615 +0.000000 +0.280830 +0.000000
-0.117131 +0.011276 -0.102307 +0.000000 -0.105249 +0.000000 -0.014098 -0.076967
+0.006400 +0.035706 +0.182779 -0.000000 +0.085177 +0.000000 -0.036445 -0.000000
-0.127696 -0.106880 +0.266063 -0.000000 -0.281029 +0.000000 +0.000926 -0.208331
+0.013443 -0.037179 +0.052453 -0.000000 +0.043364 -0.000000 -0.260687 -0.000000
-0.023556 +0.076327 -0.162805 +0.000000 -0.064979 +0.000000 -0.005174 -0.130230
-0.071662 +0.006476 +0.164528 -0.000000 -0.063164 +0.000000 +0.295960 +0.000000
-0.087528 +0.030215 -0.190650 +0.000000 -0.157589 +0.000000 +0.081917 -0.028998
-0.082536 +0.031928 +0.274398 -0.000000 -0.066085 +0.000000 -0.060940 -0.000000
```

- 제미나이를 활용해 생성형 AI를 실험하고, 프롬프트를 설계하고, 결과를 바로 확인할 수 있도록 만든 웹 기반 개발 환경 (a.k.a 개인 AI 실험실!)
- 홈 링크: <https://aistudio.google.com/>
- 앱 빌드 링크: <https://aistudio.google.com/apps>

Google AI Studio

- 제미나이를 활용해 생성형 AI를 실험하고, 프롬프트를 설계하고, 결과를 바로 확인할 수 있도록 만든 웹 기반 개발 환경 (a.k.a 개인 AI 실험실!)
- 홈 링크: <https://aistudio.google.com/>
- 앱 빌드 링크: <https://aistudio.google.com/apps>

Google AI Studio



Prompting and Build

The screenshot shows the Google AI Studio interface. On the left is a sidebar with navigation options: 'EXPLORE' (Playground, History), 'BUILD' (New app, My apps, Gallery), and 'MANAGE' (Dashboard, Documentation). The main area has a header 'Build your ideas with Gemini' and a large text input box. The input box contains a Korean prompt about building a web-based image classification app using a Teachable Machine model. The prompt includes a URL and a list of categories. A red checkmark is placed over the 'Build' button at the bottom right of the input box. Below the input box is a row of buttons for integrating with other Google services: 'Build an Android app', 'Google Drive', 'Google Sheets', 'Gmail', and 'Google Calendar'.

Google AI Studio

EXPLORE

- Playground
- History

BUILD

- New app
- My apps
- Gallery

MANAGE

- Dashboard
- Documentation

Build your ideas with Gemini

Google Teachable Machine으로 웹캠 기반 이미지 분류 모델을 만들었어.
모델은 Teachable Machine에서 온라인으로 업로드했고, 아래 URL을 사용할 거야.
https://teachablemachine.withgoogle.com/models/5-ImgrP...

이 모델을 이용해서 사용자가 웹캠을 켜면 실시간으로 이미지를 분류해주는 앱을 만들어줘.

현재 내 모델의 카테고리는 3개야.

["Soomin", "Cool", "Shy"]

각 카테고리의 의미는 다음과 같아.

Build

Upgrade to unlock more
Access higher limits, Pro models, and more.

Build an Android app Google Drive Google Sheets Gmail Google Calendar

Prompt

Google Teachable Machine으로 웹캠 기반 이미지 분류 모델을 만들었어.
모델은 Teachable Machine에서 온라인으로 업로드했고, 아래 URL을 사용할 거야.

[TEACHABLE_MACHINE_MODEL_URL]

이 모델을 이용해서 사용자가 웹캠을 켜면 실시간으로 이미지를 분류해주는 앱을 만들어줘.

현재 내 모델의 카테고리는 3개야.

["Soomin", "Cool", "Shy"]

각 카테고리의 의미는 다음과 같아.

```
{  
  "soomin": "여자 얼굴",  
  "Cool": "선글라스를 쓴 얼굴",  
  "Shy": "손으로 얼굴을 가린 모습"  
}
```

나중에 다른 모델을 사용할 수도 있으니 위의 [카테고리]와 {설명}은 쉽게 바꿀 수 있게 만들어줘.
각 분류 결과에 따라 짧고 위트 있는 멘트를 보여줘.

예시:

- Soomin: "Yep, you're Soomin."
- Cool: "I like your shades 😎"
- Shy: "Hide your face, hide your secrets."

Prompt

가장 높은 확률의 결과를 크게 보여주고, confidence도 같이 보여줘.

디자인은 요즘 Gen Z / Alpha Gen이 좋아할 만한 힙한 카메라 서비스처럼 만들어줘.

Capture 버튼을 만들어서 현재 웹캠 화면 + 분류 결과 + 멘트를 하나의 카드 이미지로 저장할 수 있게 해줘.
인스타그램에 공유하고 싶은 비주얼로 만들어

모바일에서도 잘 작동하게 해줘.

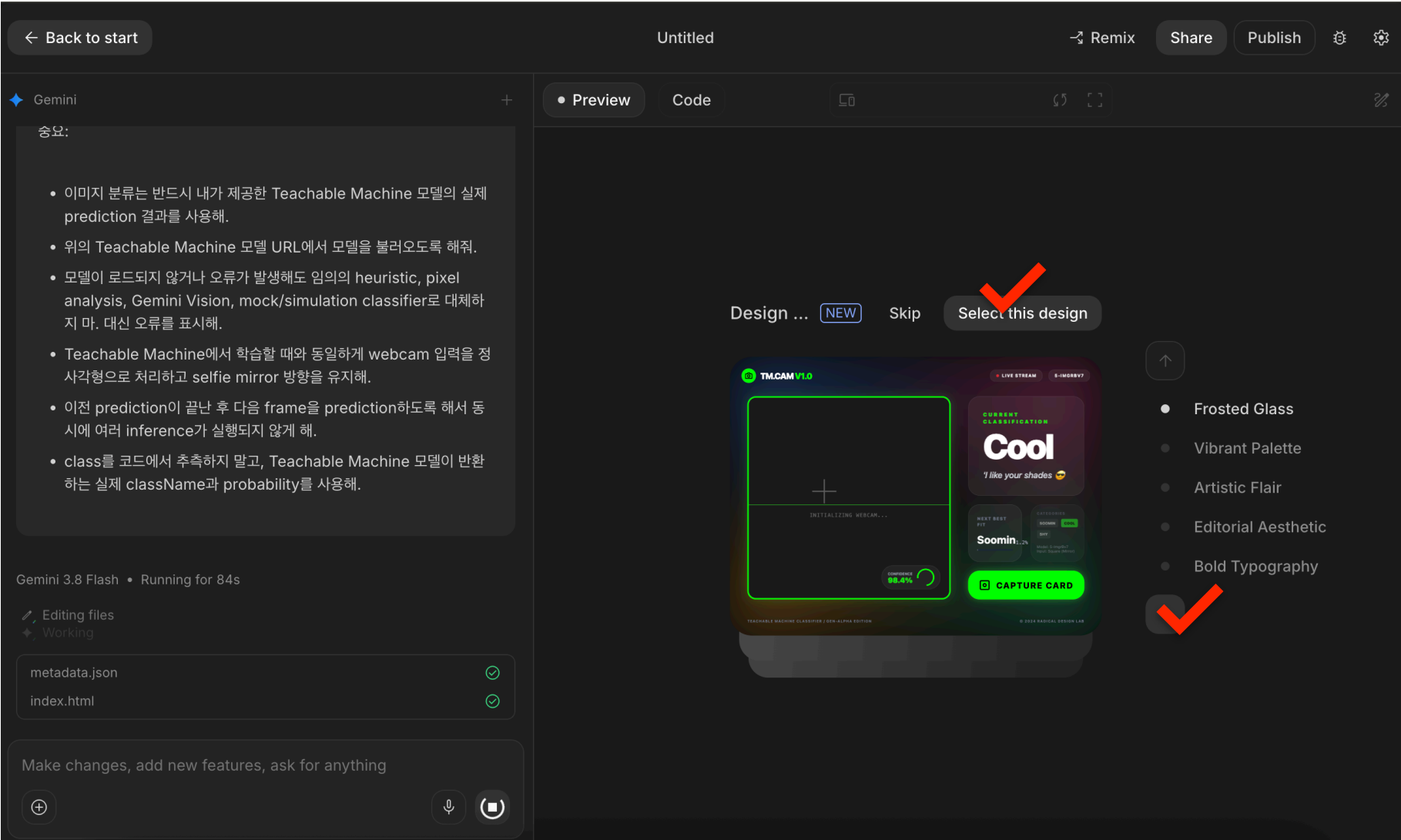
중요:

- 이미지 분류는 반드시 내가 제공한 Teachable Machine 모델의 실제 prediction 결과를 사용해.
- 위의 Teachable Machine 모델 URL에서 모델을 불러오도록 해줘.
- 모델이 로드되지 않거나 오류가 발생해도 임의의 heuristic, pixel analysis, Gemini Vision, mock/simulation classifier로 대체하지 마. 대신 오류를 표시해.
- Teachable Machine에서 학습할 때와 동일하게 webcam 입력을 정사각형으로 처리하고 selfie mirror 방향을 유지해.
- 이전 prediction이 끝난 후 다음 frame을 prediction하도록 해서 동시에 여러 inference가 실행되지 않게 해.
- class를 코드에서 추측하지 말고, Teachable Machine 모델이 반환하는 실제 className과 probability를 사용해.

에러 방지

- 라이브러리는 tfjs@1.3.1, @teachablemachine/image@0.8.5 버전으로 고정해. latest는 쓰지 마.
- 모델은 앱 전체에서 딱 1회만 로드해. 콜백 함수는 useRef나 useCallback으로 고정해서 재로딩 루프가 생기지 않게 해.
- video에 muted, playsinline을 넣고 play() 같은 Promise는 전부 catch로 감싸. 효과음이나 AudioContext는 넣지 마.
- 카메라 권한 거부 / 모델 URL 로드 실패 / HTTPS 아님, 이 3가지는 각각 한국어 안내 문구를 화면에 띄워줘. 그리고 에러가 나도 흰 화면이 되지 않게 ErrorBoundary를 넣어줘.

Select the Design



Fix the errors

← Back to start

Teachable Machine Webcam Classifier

Remix Share Publish

◆ Gemini

◆ Gemini 3.8 Flash • Ran for 192s

accents, an explanatory prompt callout, and a live stream status pulse.

• Top HUD & Live Digital Clock: Integrated status chips (60 FPS, 224×224 input, on-device TFJS) alongside a live-ticking timestamp clock matching the theme format.

• Viewfinder & Floating Cards: Styled the camera frame with rounded 40px borders and deep zinc framing (border-[6px] border-zinc-800), housing the top-left prediction badge with colored square emoji containers and the bottom-right confidence card displaying real-time probabilities and witty quotes.

• Action Log Bar & Shutter: Added the bottom log bar tracking model detections paired with a high-contrast brutalist CAPTURE button (hover:bg-[#00FF66]) to trigger snapshots and photocard generation.

Checkpoint View changes Restore

1 error running the code Fix

◆ AI Features Add sound effects Confetti effect

Make changes, add new features, ask for anything

• Preview Code

Dismiss all 1 1 8

App Debug [vite] connecting...

App Debug [vite] connected.

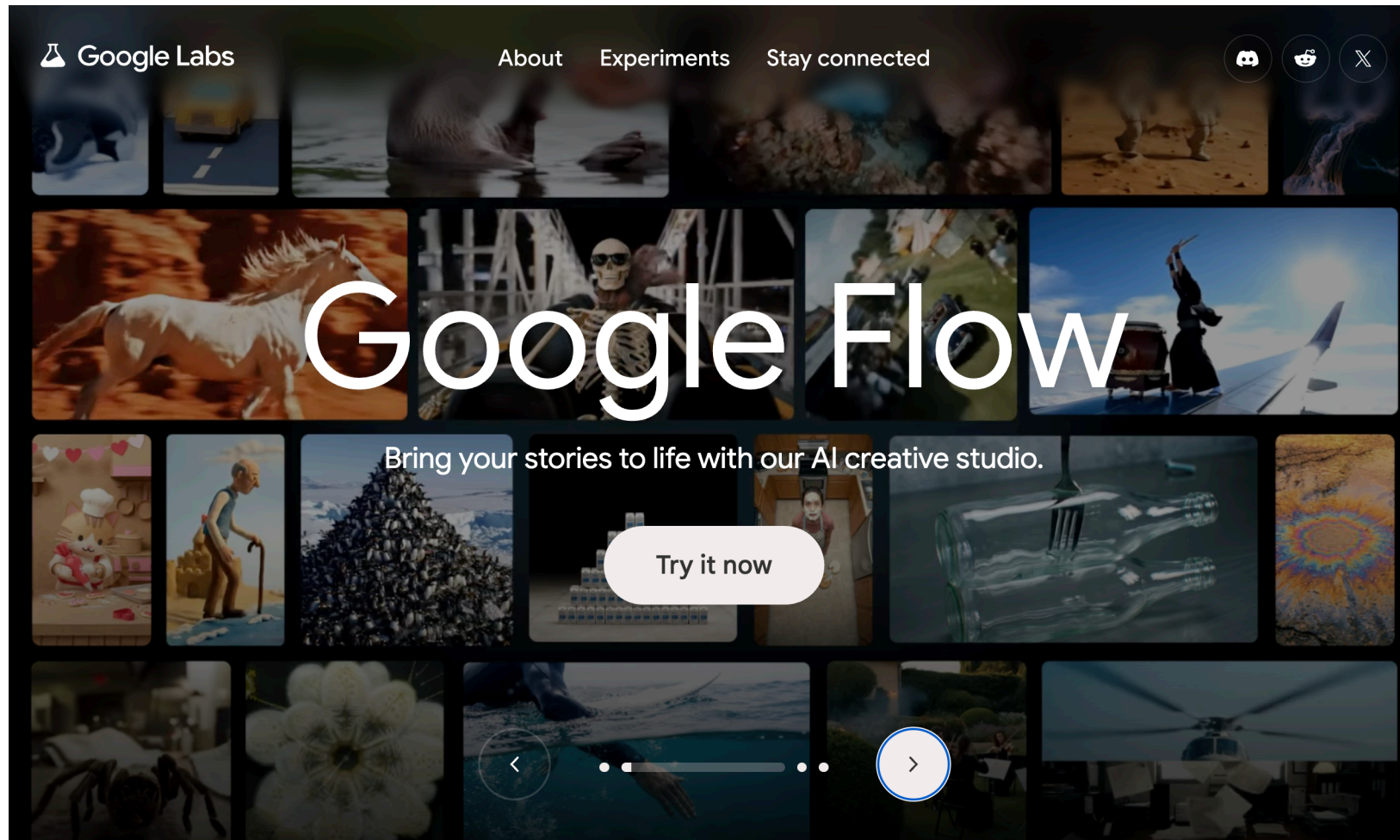
App Error Uncaught

App Warning %s

%s

An error occurred in the <div> component. Consider adding an error boundary to your tree to customize error handling behavior. Visit <https://react.dev/link/error-boundaries> to learn more about error boundaries.

참고 labs.google



코드 다운로드 받기

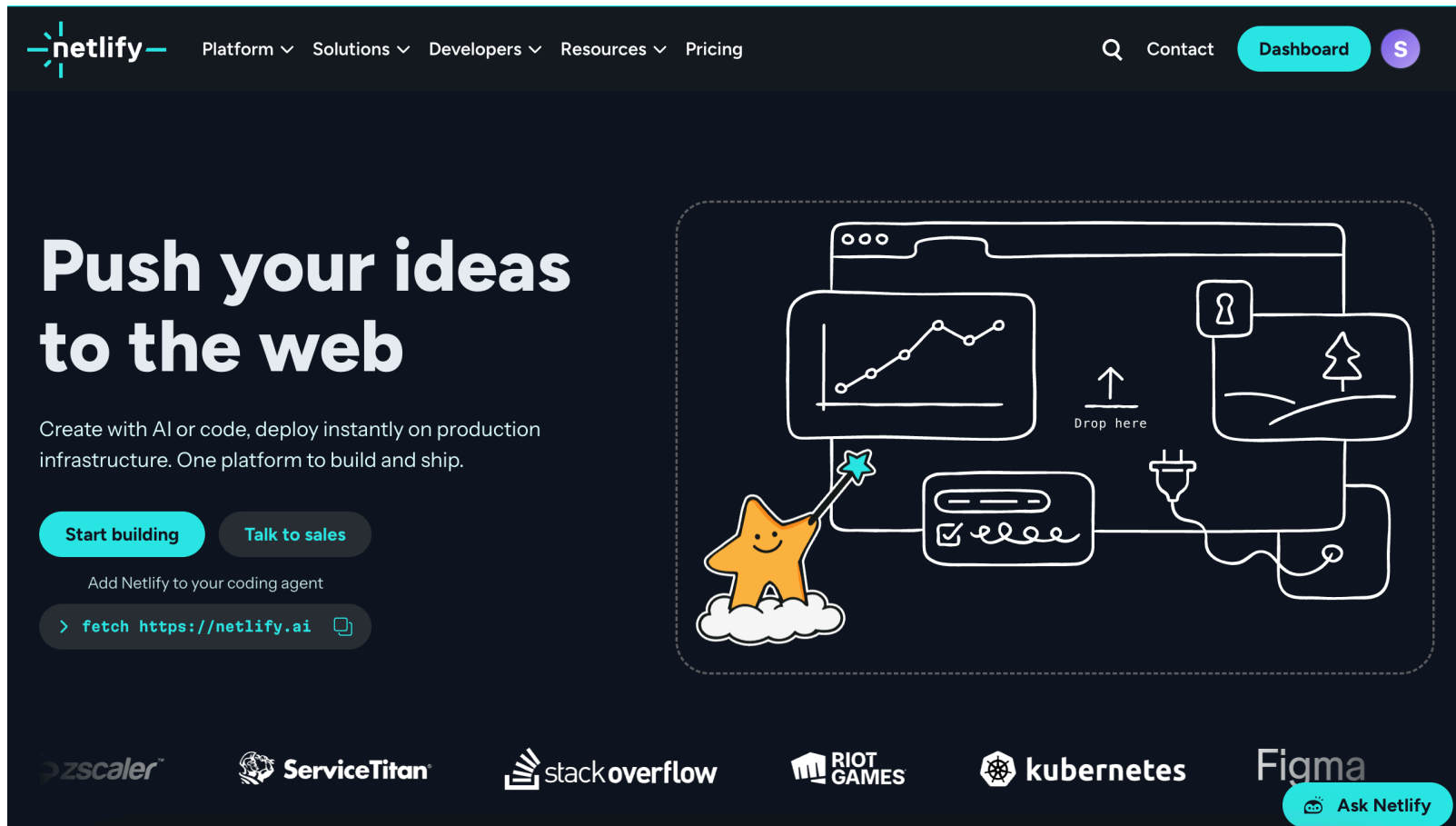
The screenshot shows the Teachable Machine Webcam Classifier interface. The 'Code' tab is selected, displaying the source code for the application. A red checkmark is placed over the 'Code' tab label. A red box with the text '코드 선택' (Code Selection) is overlaid on the left side of the code editor. Another red box with the text '다운로드 zip file' (Download zip file) is overlaid on the right side of the code editor, pointing to the 'Download as .zip file' button. The interface includes a 'Back to start' button, 'Remix', 'Share', and 'Publish' buttons. The code editor shows the following code:

```
1 /**
2  * @license
3  * SPDX-License-Identifier: Apache-2.0
4  */
5
6 import React, { useState, useMemo, useEffect, use
7 import { Sliders, HelpCircle, Eye, ShieldCheck, Camera, Sparkles } from 'lucide-
8 import { PredictionResult, CategoryMeta,
9 import { DEFAULT_MODEL_URL, DEFAULT_CATEG
10 import { CameraView } from './components,
11 import { CategoryProbabilities } from './
12 import { CaptureModal } from './component
13 import { SettingsModal } from './componen
14
15 export default function App() {
16   // Config state persisted in localStorage
17   const [modelUrl, setModelUrl] = useState<string>(() => {
18     try {
19       return localStorage.getItem('tm_model_url') || DEFAULT_MODEL_URL;
20     } catch {
21       return DEFAULT_MODEL_URL;
22     }
23   });
24
25   const [categories, setCategories] = useState<Record<string, CategoryMeta>>(()
26     try {
27       const saved = localStorage.getItem('tm_categories');
28       if (saved) return JSON.parse(saved);
29     } catch {
30       // ignore
31     }
32     return DEFAULT_CATEGORIES;
33   });
```

The interface also includes a 'Checkpoint' section with a 'View changes' button and a 'Restore' button. At the bottom, there is a search bar with the text 'Make changes, add new features, ask for anything' and a microphone icon.

Netlify 네트리파이

- 웹 프로젝트의 빌드, 배포, 운영을 자동화하는 웹 개발 플랫폼
- 링크: <https://www.netlify.com/>



Netlify 네트리파이

The image shows the Netlify dashboard interface. On the left is a dark sidebar with a search bar and a list of navigation items: Projects, Builds, Extensions, Members, Audit log, Security Scorecard, Usage & billing, and Team settings. The 'Projects' item is highlighted with a red box and a red arrow pointing to it from a red text box. The main content area is titled 'Projects' and contains a search bar, an 'Add new project' button, and a list of projects. One project, 'my-face-check', is visible. Below the project list is a large dashed box with the text: 'Want to deploy a new project? Drag and drop your folder here. Drag and drop your project folder, zip file, or a single HTML file to deploy instantly. Or [browse files to upload](#) or [choose a folder](#).' A red arrow points from a red text box to this dashed box. To the right of the dashed box is a blue folder icon, with a red dashed arrow pointing from the text box to it. At the bottom of the sidebar is an 'Upgrade' button and the user's name 'Soomin Kim' with their email. The bottom of the main area has links for Docs, Pricing, Support, Blog, Changelog, and Terms.

Project 메뉴 클릭 후

아래 영역에 다운로드 받은 폴더 드래그 앤 드롭

Want to deploy a new project? Drag and drop your folder here.
Drag and drop your project folder, zip file, or a single HTML file to deploy instantly.
Or [browse files to upload](#) or [choose a folder](#).

Netlify 네트리파이

The screenshot shows the Netlify dashboard interface. On the left is a sidebar with navigation icons and labels: Projects, Project overview, Project configuration, Deploys (highlighted with a blue diamond), Agent runs, Environment variables, Logs & metrics, Web security, Domain management, Forms, Identity, and Data & Storage. The main content area is titled 'Published deploy for serene-hummingbird-f1f964'. It shows the deployment was built from a drop deployment and occurred today at 6:09 PM. There are buttons for 'Open production deploy', 'Lock to stop auto publishing', and 'Options'. A large red banner with white Korean text '업로드 완료!' (Upload complete!) is overlaid on the 'Deploy summary' section. The summary shows '3 new files uploaded' (1 generated page and 2 assets changed) and 'Build time: 15s. Total deploy time: 14s'. At the bottom, there is a 'Deploy log' section with a 'Preview' button and icons for copy, up, down, and maximize log. On the right side of the dashboard, there is a 'Fancy a game?' section with a 'Play game' button.

Projects / serene-hummingbird-f1f964

News Support

Project overview
Project configuration
◆ Deploys
Agent runs
Environment variables
Logs & metrics
Web security
Domain management
Forms
Identity
Data & Storage

Published deploy for serene-hummingbird-f1f964
Build from drop deployment
Today at 6:09 PM
Permalink

Open production deploy Lock to stop auto publishing
Options

업로드 완료!

Deploy summary

- 3 new files uploaded
1 generated page and 2 assets changed.
- Build time: 15s. Total deploy time: 14s
Build started at 6:09:09 PM and ended at 6:09:24 PM. [Learn more about build minutes](#)

Deploy log Preview Copy Up Down Maximize log

Fancy a game?
This deploy is done, but hey, you can still play if you want to!
Play game

Netlify 네트리파이

The screenshot shows the Netlify dashboard for a project named 'serene-hummingbird-f1f964'. The interface is in dark mode. On the left, a sidebar contains navigation links: 'Project overview' (highlighted with a red box and an arrow), 'Project configuration', 'Deploys', 'Agent run', 'Logs & metrics', 'Web security', 'Domain management', 'Forms', 'Identity', and 'Data & Storage'. The main content area shows the project name, a 'Private' status, and the URL 'serene-hummingbird-f1f964.netlify.app' (highlighted with a red box and an arrow). Below this, it says 'Last deployed from Netlify Drop' and 'Published at 6:09 PM.' A red box with white text '녹색 링크 클릭! (카메라는 사용 허용)' points to the URL. On the right, there's a section 'Protect and secure access to your project' with a 'Upgrade to protect' button. At the bottom, there's a 'Production deploys' section with a drag-and-drop area. A red box with white text 'Project Overview' is also present on the left side of the main content area.

Projects / serene-hummingbird-f1f964

News Support

Project overview

Project configuration

Deploys

Agent run

Project Overview

serene-hummingbird-f1f964 Private

serene-hummingbird-f1f964.netlify.app

Last deployed from Netlify Drop

Published at 6:09 PM.

녹색 링크 클릭!
(카메라는 사용 허용)

Project configuration

Work with an AI agent New

Ask an agent to create a new page, build a dashboard, update an article, etc.
Review, iterate and share.

+ OpenAI Codex (Auto) Build

The agent publishes changes in a preview - your live site won't change. [Learn more](#)

Production deploys

Drag and drop your project folder here to deploy new changes

Protect and secure access to your project

Customize how visitors access your project using password protection available on the Pro plan. [Learn more](#)

Upgrade to protect

Observability

No data in the last hour

Netlify 네트리파이

The image shows the Netlify dashboard for a project named 'serene-hummingbird-f1f964'. The project is currently 'Private'. A red arrow points to the 'Make public' button, which is highlighted with a red box containing the text 'URL을 Public 하게 만들기' (Make URL public). Another red arrow points to the 'Make public' modal, which is also highlighted with a red box containing the text '원하는 URL 입력' (Enter desired URL). The modal shows the 'Website URL' field with the value 'my-face-vibe' and a dropdown menu set to 'Public' (Anyone on the internet). A red checkmark is visible at the bottom right of the modal.

Projects / serene-hummingbird-f1f964

Project overview

Project configuration

Deploys

Agent runs

Environment variables

Logs & metrics

Web security

Domain management

Forms

Identity

Data & Storage

serene-hummingbird-f1f964 Private

serene-hummingbird-f1f964.netlify.app

Last deployed from Netlify Drop

Published at 6:09 PM.

Make public Project configuration Share

URL을 Public 하게 만들기

Work with an AI agent New

Prompt templates

Review, iterate and share.

+ OpenAI Codex (Auto) Build

The agent publishes changes in a preview - your live site won't change. [Learn more](#)

Production deploys

Drag and drop your project folder here to deploy new changes

Protect and secure access to your project

Customize how visitors access your project using password protection available on the Pr

원하는 URL 입력

Make public

Website URL

my-face-vibe .netlify.app

Project name is available

When you go live, who can access your project?

Public Anyone on the internet

Deploy previews remain private to your team.

Make public Cancel

Want to keep it private and invite people? **Share**